Multiyear Sentinel-1/2 and CHIRPS Integration for Flood Prediction and Agricultural Exposure Assessment in the Senegal River Valley

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Abstract

Floods threaten agricultural production in the Senegal River Valley, yet flood mapping, crop exposure assessment, and early prediction are often treated as separate tasks. We analyze the available 2019–2024 Google Earth Engine exports, which combine Sentinel-1, Sentinel-2, CHIRPS, historical water indicators, and 200 repeated sampling locations. The point–period table contains 19,200 records; 14,519 have a valid current Sentinel-1-derived flood label and 13,583 have a valid next-period target, with 3.76% positive labels. The six annual area tables each contain 16 periods; mean mapped inundated area per period ranges from 128,055 to 212,994 ha. A chronological baseline experiment trained on 2019–2023 and tested on 2024 achieved ROC-AUC 0.981 and average precision 0.768 for a class-balanced Random Forest at a fixed 0.5 threshold. These are preliminary results against a satellite-derived proxy, with repeated sites shared across training and test years. The available agricultural-exposure rasters are color-rendered images, not class-valued masks, and no field damage or yield observations are provided. Therefore, crop exposure can be discussed only as a mapping objective; measured crop damage and production loss cannot be inferred from these exports.

Keywords: Flood prediction, agricultural exposure, Sentinel-1, Sentinel-2, CHIRPS, Dynamic World, Google Earth Engine, Senegal River Valley

1 Introduction

Floods can disrupt agricultural production through direct crop submergence, delayed field operations, erosion, and prolonged soil saturation. In semi-arid agricultural regions, however, the relationship between inundation and crop damage depends on flood duration, crop type, phenological stage, and local water management. Consequently, mapping flooded pixels alone does not quantify agricultural exposure, and agricultural exposure does not by itself prove crop damage or yield loss.

Recent Earth-observation studies combine synthetic aperture radar (SAR), optical satellite data, vegetation time series, and machine learning to map inundation or estimate crop damage. Deep-learning approaches such as U-Net have performed well for flood segmentation, while Random Forest (RF) has been used for field-level damage classification and regression. A recent phenology-guided study reported pixel-level RF coefficients of determination of 0.62 using early observations and 0.77 using later observations for a vegetation-index-based proxy of crop damage [**?**]. A separate field-level study using agronomist observations reported an overall accuracy of 0.74 for three agricultural damage classes [**?**]. These metrics concern different targets and should not be compared as if they measured the same task.

Despite this progress, three gaps remain relevant to the Senegal River Valley: (i) the need to examine flood dynamics across multiple seasons rather than a single season; (ii) the need to quantify the spatial overlap between potential inundation and candidate cropland; and (iii) the need for temporally honest prediction evaluation that separates earlier seasons from a later test season. This study addresses these gaps through a reproducible Google Earth Engine (GEE) workflow.

The objectives are to:

1. construct a multiyear, multimodal point–time dataset from Sentinel-1, Sentinel-2, CHIRPS, and historical water information;
2. map potential newly inundated surfaces and summarize their area and temporal frequency;
3. estimate the area of candidate agricultural land exposed to potential inundation;
4. prepare a next-period inundation target and define a chronological model evaluation strategy.

The term *agricultural exposure* is used deliberately. Without independent agronomic or yield observations, the workflow cannot establish that an exposed field suffered a particular level of damage or production loss.

2 Related work

2.1 Flood mapping from radar and optical imagery

Sentinel-1 SAR is useful for flood screening because radar observations can be acquired under cloud cover. Threshold-based methods are computationally simple, but a single backscatter threshold can confuse water with smooth bare soil, radar shadow, or other low-backscatter surfaces. Comparative work across several flood events evaluated RF,

SVM, U-Net, DeepLabV3, and Mask R-CNN; U-Net achieved an intersection-over-union score of 0.763 in the reported experiment [**?**]. This result supports the use of segmentation models when suitable training labels and computational resources are available, while also emphasizing the importance of cross-event validation.

2.2 Agricultural exposure and crop damage

A 2026 study estimated crop damage under long-duration inundation using multi-source SAR–optical imagery and a phenology-guided vegetation-index loss proxy. Its RF regression obtained $R^2 = 0.62$ with early-period features and $R^2 = 0.77$ with later-period features [**?**]. Because the target was derived from EVI time-series deviation, it should be interpreted as a proxy for growth suppression rather than direct measured yield loss. In another 2026 study, RF classified three damage levels using Sentinel-2 indices, topography, and flood extent; the reported overall accuracy and F1-score were 0.74 and 0.74, respectively, based on 412 fields with in-situ assessments [**?**]. These studies illustrate both the potential and the continuing need for field-based validation.

An integrated study in Hadejia, Nigeria, combined satellite time series and participatory mapping to assess crop damage and recovery, reporting flood mapping and crop-recovery accuracy above 90% [**?**]. The results are context-specific and cannot be transferred directly to another river valley without validation. More generally, differences in labels, spatial units, validation design, and metrics make cross-paper performance comparisons indicative rather than definitive.

3 Methods

3.1 Study area, period, and sampling framework

The study focuses on the Senegal River Valley (SRV), Senegal. A predefined Earth Engine feature collection, `projects/project37246/assets/points_200_vallee`, provides 200 sampling locations with unique identifiers and geographic coordinates. The multiyear analysis covers the growing-season window from 1 June to 31 October for each year from 2019 to 2024. Sixteen consecutive ten-day periods are generated per year. If all locations and periods are retained, the point–time dataset contains up to $200 \times 16 \times 6 = 19{,}200$ records. The final number of usable records depends on satellite coverage and valid-pixel availability.

The primary key is (`point_id`, `date_debut`). Dates and identifiers are used for joins; floating-point coordinate equality is not used as the primary matching criterion.

3.2 Sentinel-1 observations and potential inundation

Sentinel-1 GRD imagery is obtained from `COPERNICUS/S1_GRD`. Images in IW mode with VV and VH polarizations are filtered to the study region and each ten-day period. Median composites are generated, and a spatial median filter is applied to reduce

speckle. The VV backscatter is used for an initial potential-water screen:

$$W_{i,t} = \begin{cases} 1, & VV_{i,t} < \tau_{\mathrm{VV}}, \\ 0, & VV_{i,t} \geq \tau_{\mathrm{VV}}, \end{cases} \tag{1}$$

where the initial threshold is $\tau_{\mathrm{VV}} = -16$ dB. This is an operational screening threshold, not a universally valid water/land boundary, and it should be tested against independent reference data.

For each year, a pre-season reference is generated from 1 March to 15 May. Let $W_{i,y}^{\mathrm{ref}}$ be the potential-water indicator derived from that reference. Potential newly inundated pixels are defined as

$$F_{i,t} = W_{i,t}\left(1 - W_{i,y}^{\mathrm{ref}}\right). \tag{2}$$

The label $F_{i,t} = 1$ therefore indicates water detected in the current composite but not in the pre-season reference. It is not equivalent to independently confirmed flooding. Pixels without valid radar observations are treated as missing rather than as non-flooded.

The JRC Global Surface Water product (`JRC/GSW1_4/GlobalSurfaceWater`) provides historical water occurrence and maximum extent. These layers are contextual covariates and are not interpreted as event-specific flood observations.

3.3 Sentinel-2 vegetation and water indices

Sentinel-2 Level-2A imagery is obtained from `COPERNICUS/S2_SR_HARMONIZED`. Scenes with reported cloud percentage below 40% are retained, and the scene classification layer is used to mask cloud, shadow, defective, and no-data classes. Ten-day median composites are used to calculate NDVI, EVI, MSAVI, and NDWI:

$$NDVI = \frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + \rho_{B4}}, \tag{3}$$

$$EVI = 2.5\frac{\rho_{B8} - \rho_{B4}}{\rho_{B8} + 6\rho_{B4} - 7.5\rho_{B2} + 1}, \tag{4}$$

$$MSAVI = \frac{2\rho_{B8} + 1 - \sqrt{(2\rho_{B8} + 1)^2 - 8(\rho_{B8} - \rho_{B4})}}{2}, \tag{5}$$

$$NDWI = \frac{\rho_{B3} - \rho_{B8}}{\rho_{B3} + \rho_{B8}}. \tag{6}$$

Here, ρ_{B2}, ρ_{B3}, ρ_{B4}, and ρ_{B8} denote scaled blue, green, red, and near-infrared reflectance. NDWI is treated as a complementary water-related indicator rather than definitive evidence of inundation. Optical missingness caused by clouds or absent acquisitions is preserved.

3.4 CHIRPS precipitation and antecedent rainfall

Daily precipitation is obtained from `UCSB-CHG/CHIRPS/DAILY`. For each point and ten-day period, rainfall is aggregated to produce the ten-day total, maximum daily rainfall, number of rainy days, and antecedent accumulations such as 30-day and 90-day rainfall. For a window of k days:

$$P_{i,t}^{(k)} = \sum_{d=t-k+1}^{t} P_{i,d}, \qquad k \in \{30, 90\}. \tag{7}$$

A standardized precipitation index should only be reported if it is computed using a documented accumulation scale, reference period, and suitable statistical distribution. A simple standardized rainfall anomaly should not be described as a fully fitted SPI.

3.5 Candidate agricultural mask

Candidate agricultural pixels are derived from the `crops` class-probability band in Google Dynamic World (`GOOGLE/DYNAMICWORLD/V1`). For each year, the seasonal mean crop probability is calculated over 1 June–1 November. Pixels with mean crop probability at or above 0.30 are retained as a provisional agricultural mask:

$$C_{p,y} = \mathbb{I}\left(\overline{q}_{p,y}^{\text{crop}} \geq 0.30\right), \tag{8}$$

where $\overline{q}_{p,y}^{\text{crop}}$ is the seasonal mean probability and $\mathbb{I}$ is the indicator function. The threshold is a configurable starting value and must be assessed for sensitivity. Dynamic World probabilities indicate likely land-cover class, not crop species, crop calendar, or verified parcel boundaries. Agricultural exposure estimates are therefore conditional on the mask and its classification accuracy.

3.6 Area and temporal frequency of potential inundation

For a binary inundation mask $F_{p,t}$ and pixel area a_p in square metres, total potential inundated area for period t is calculated as

$$A_t^{\text{flood}} = \frac{1}{10^4} \sum_p a_p F_{p,t}, \tag{9}$$

where only valid radar pixels are included. The potential agricultural area exposed is

$$A_t^{\text{crop,flood}} = \frac{1}{10^4} \sum_p a_p F_{p,t} C_{p,y}. \tag{10}$$

The denominator for the exposed-cropland percentage includes only candidate crop pixels observed by valid radar in the corresponding period:

$$R_t^{\mathrm{crop}} = 100\,\frac{A_t^{\mathrm{crop,flood}}}{A_t^{\mathrm{crop,observed}}}. \tag{11}$$

A missing radar pixel is not counted as a non-flooded pixel. Seasonal frequency maps are computed from the number of valid periods with potential inundation divided by the number of valid radar observations at each pixel. Crop-exposure frequency uses the periods for which both the candidate crop mask and radar observation are valid. Connected-component counts, if reported, are treated as approximate and checked for sensitivity to spatial scale and component-size limits.

3.7 Next-period prediction target and leakage control

To prepare a forecasting dataset, the candidate target is the potential-inundation indicator for the next ten-day period:

$$Y_{i,t+1} = F_{i,t+1}. \tag{12}$$

The last period of each season has no next-period label and is excluded from supervised evaluation. Predictors must be restricted to information available at the forecast origin. Contemporaneous or future Sentinel-1 variables used to construct the target must not be included as unrestricted predictors in a model that claims to forecast inundation independently. Sentinel-2, rainfall, antecedent rainfall, and historical water context can be considered as predictors subject to timestamp checks and missing-data analysis.

The proposed chronological evaluation uses 2019–2023 for model development and 2024 as a held-out test season. Hyperparameter selection is performed only within the training years, preferably using year-blocked validation. If independent flood labels are unavailable, the task must be described as prediction of Sentinel-1-derived potential inundation, not prediction of independently verified floods.

3.8 Model candidates and evaluation

The local analysis script evaluates three tabular classifiers for $Y_{i,t+1}$: a prior-prevalence dummy classifier, class-balanced logistic regression, and a class-balanced Random Forest with 300 trees and a minimum leaf size of five. The 29 predictors comprise coordinates, calendar variables, current-period radar and optical measurements, historical-water indicators, rainfall summaries, and the radar-validity flag. The entirely missing `NDVI_ref`, current flood label, next-period target, and identifiers are excluded. Remaining predictor missingness is imputed with training-set medians; the logistic model also standardizes predictors. The split is chronological (2019–2023 train; 2024 test), and reported classification metrics use a fixed probability threshold of 0.5. ROC-AUC, average precision, Brier score, precision, recall, F1-score, and balanced accuracy are reported. This single-year evaluation is exploratory: the same 200 sites occur in

training and test data, so it does not measure transfer to unseen locations. A spatially blocked, year-blocked nested validation and independent flood reference labels are required before operational claims.

For future work, boosted trees (CatBoost, XGBoost, or LightGBM) are plausible tabular candidates, but they should be compared only under the same blocked validation. Flood recurrence should be modeled as a count with the number of valid observations as exposure (e.g., a Poisson or negative-binomial model, with a hurdle formulation if excess zeros remain). Event-level area prediction has only 96 annual-period records in the current exports and is too small for a high-capacity learner. Cropland exposure requires a numeric, validated crop mask aligned with flood masks; damage or yield-loss prediction additionally requires field damage classes, crop type/phenology, or independent yield references. Image segmentation (e.g., U-Net) should be considered only after assembling independently labeled flood and crop pixels. These are proposed extensions, not results of the present experiment.

3.9 Validation and uncertainty

The VV threshold, dry-reference dates, Dynamic World crop-probability threshold, and spatial scale can influence estimated areas. Sensitivity analysis should vary these settings and compare mapped outputs with independent flood references where available. Spatially nearby points are not independent replicates; where feasible, validation should additionally group locations spatially. Cloud gaps, missing radar acquisitions, mixed pixels, seasonal water, irrigation, and land-cover classification errors are documented as sources of uncertainty.

4 Results

4.1 Dataset completeness and temporal coverage

The main export contains 19,200 rows for 200 sites, six years, and 16 ten-day periods per year. No duplicate (`point_id`, `date_debut`) keys were found. There are 14,519 valid current flood labels (3.73% positive) and 13,583 valid next-period targets (3.76% positive); 29.26% of next-period labels are missing. Sentinel-1 fields and current labels are missing in 24.38% of rows, optical indices in 26.83%, and CHIRPS summaries in 0.50–1.16%. The `NDVI_ref` field is entirely missing and is excluded from modeling. The next-period target was checked against a within-year, within-site one-period shift of the current flood label; all 13,583 comparable labels aligned.

4.2 Multiyear inundation dynamics

Across the 96 exported annual-period records, mean mapped area per period ranged from 128,055 ha in 2024 to 212,994 ha in 2019. The largest period-level estimate was 397,106 ha in 2021. The annual sum of these 16 period estimates is reported in the generated CSV as hectares-periods; it is not a unique seasonal footprint because the same land can be counted in multiple periods. Target prevalence varied from 3.47% in 2021 to 4.41% in 2019. Across all valid current labels, the mean sampled-point inundation frequency was 3.74%; the median site frequency was zero, while a few sites

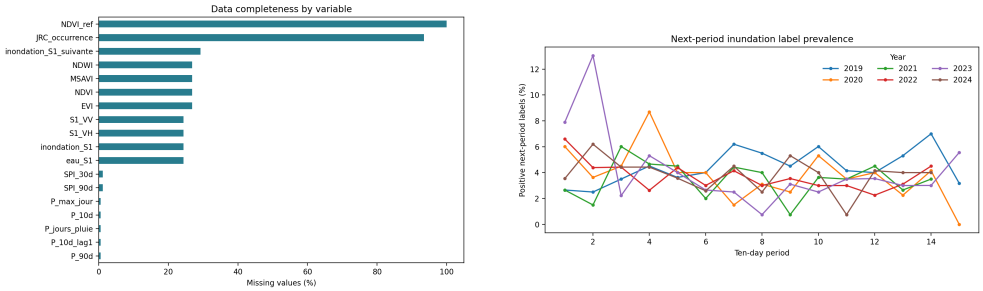


Fig. 1 Data missingness among the most incomplete variables (left) and observed prevalence of the next-period inundation target by ten-day period and year (right). The prevalence denominator includes only rows with a valid target.

Table 1 Period-level mapped inundated area and next-period target prevalence by year. Area values are not unique seasonal totals.

Year	Valid target rows	Positive targets (%)	Mean area per period (ha)	Maximum period area (ha)
2019	2,517	4.41	212,994	339,014
2020	2,337	3.72	148,678	300,576
2021	2,391	3.47	181,281	397,106
2022	2,111	3.60	142,964	220,249
2023	2,115	3.50	128,213	190,107
2024	2,112	3.79	128,055	276,264

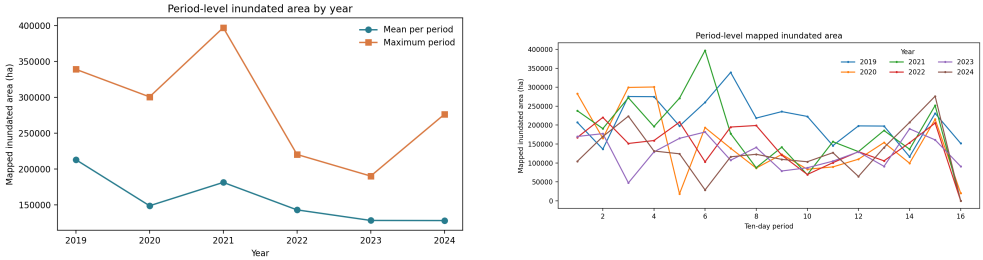


Fig. 2 Area summaries derived from the six period-level CSV exports: annual mean and maximum (left), and ten-day trajectories (right). These are mapped potential-inundation estimates, not field-validated flood extents.

had much higher recurrence. Frequencies are conditional on valid observations and the fixed sample, not unbiased estimates for every location in the valley.

4.3 Agricultural exposure

The supplied directory contains color-rendered GeoTIFF mosaics for inundation and agricultural exposure, but no numeric crop-exposure CSV or class-valued raster and no legend that permits a reproducible conversion of display colors to area. Consequently, agricultural area exposed and its percentage cannot be quantified from the available

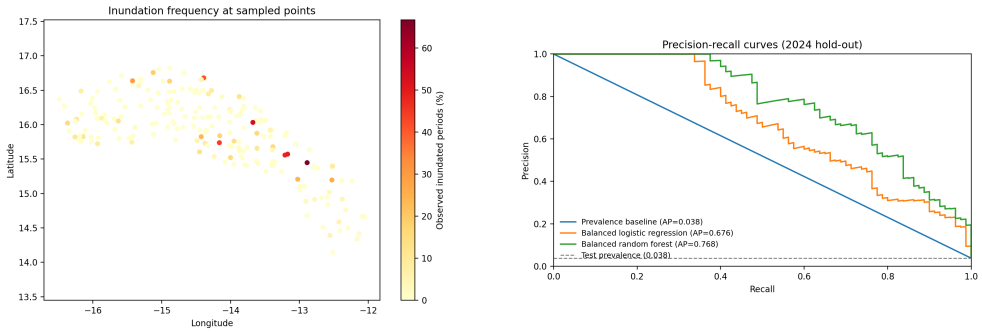


Fig. 3 Observed recurrence at the 200 sampled sites (left; valid periods only) and precision–recall curves for the held-out 2024 next-period classification task (right).

Table 2 Exploratory model performance on the 2024 chronological hold-out set. The classification threshold is 0.5; average precision is the principal ranking metric under class imbalance.

Model	Precision	Recall	F1	Bal. acc.	ROC-AUC	Avg. precision	Brier
Prevalence baseline	0.000	0.000	0.000	0.500	0.500	0.038	0.036
Balanced logistic regression	0.229	0.938	0.369	0.907	0.968	0.676	0.076
Balanced Random Forest	0.774	0.600	0.676	0.797	0.981	0.768	0.021

exports in this analysis. The point-level frequency map in Fig. **??** shows sampled flood recurrence only; it is not a cropland or damage map. No crop damage, yield reduction, or livelihood impact is measured here.

4.4 Prediction performance

Training used 11,471 labeled records from 2019–2023; the 2024 test set contained 2,112 records, including 80 positive targets. At a fixed threshold of 0.5, the Random Forest achieved precision 0.774, recall 0.600, F1 0.676, balanced accuracy 0.797, ROC-AUC 0.981, average precision 0.768, and Brier score 0.021. Logistic regression achieved ROC-AUC 0.968 and average precision 0.676, with recall 0.938 but lower precision (0.229) at the same threshold. The prevalence-only baseline had ROC-AUC 0.500 and average precision 0.038, equal to test prevalence. Class ranking is promising in this split, but calibration and generalization remain unverified; the Random Forest's Gini importance was highest for the dry-reference water indicator, followed by current radar backscatter and historical-water extent.

5 Discussion

The results show that these exports support a reproducible point-level recurrence analysis and an initial next-period classification experiment. The Random Forest ranked the derived labels well in 2024, but the test year contains only 80 positive cases, the same sample sites occur in both partitions, and the target itself is derived from the

same satellite workflow. Spatially blocked evaluation and independent flood delineations are therefore essential before interpreting this as operational prediction or transfer to new locations. The area tables summarize mapped area at each ten-day period; summing them across a season double-counts repeatedly inundated land. The supplied rendered exposure mosaics do not provide auditable crop-area measurements, and no yield or field damage outcomes are available. Results must not be interpreted as measured agricultural loss or causal crop impacts.

6 Limitations

The current workflow uses a fixed VV threshold and a pre-season water reference, which may misclassify smooth bare soil, permanent or seasonal water, and water under vegetation. Dynamic World provides class probabilities rather than field-verified crop parcels. The 200 point locations are a sampling framework and cannot, by themselves, support unbiased area estimation for the entire valley unless the sampling design and spatial extrapolation are justified. Missing sensor observations reduce the labeled sample and may not be random. The held-out 2024 year shares sites with training years, so the reported metrics do not establish spatial generalization; only one year is held out and no confidence intervals or independent flood reference are available. The raster exports are rendered RGB products, and the CSV exports do not contain a numeric agricultural-exposure measurement. Most importantly, overlap between potential inundation and candidate cropland estimates exposure, not damage severity or yield loss. Such claims require independent field surveys, crop-specific yield data, or validated crop-growth indicators.

7 Conclusion

This analysis establishes a reproducible multiyear workflow for inspecting Sentinel-1-derived inundation indicators, Sentinel-2 vegetation and water indices, CHIRPS rainfall, historical water context, and point-level recurrence in the Senegal River Valley. The 2024 hold-out experiment suggests that tabular models can rank next-period satellite-derived labels, but its single-year, spatially repeated design is not sufficient for operational claims. Numeric crop exposure and agricultural damage cannot be estimated from the currently available exports. Future work should prioritize independent flood labels, numeric validated crop masks, spatially and temporally blocked evaluation, and field or yield observations before interpreting exposure as crop damage or production loss.

Declarations

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Competing interests

The author declares no competing interests.

Data availability

The analysis used the supplied 2019–2024 CSV exports. Derived tables, model predictions, and figures are saved in the accompanying `resultats_inondations` directory. The source satellite products and Earth Engine assets remain subject to their respective access and sharing conditions; a persistent repository link should be added before submission.

Code availability

The local exploratory analysis and baseline-model script is `analyse_exploratoire_modeles.py`. It reads the CSVs from the parent data directory and regenerates the analysis outputs. The original GEE workflow and a versioned public repository should be supplied before submission.

Author contributions

To be completed according to the actual contributions of the author(s).